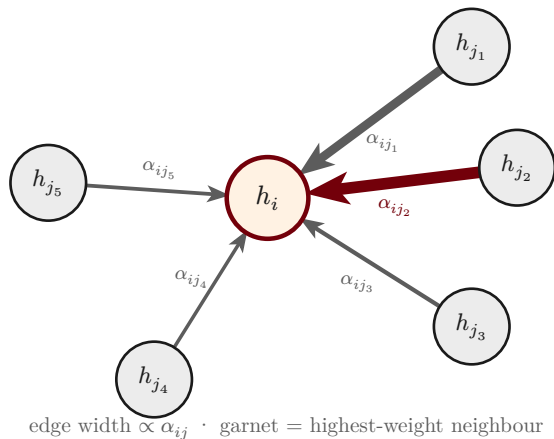


(a) attention-weighted aggregation



(b) computing α_{ij}



linear map: $z_k = Wh_k$

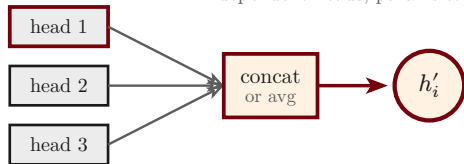
edge score: $e_{ij} = \text{LeakyReLU}(a^\top [z_i \parallel z_j])$

normalise (softmax over $j \in \mathcal{N}(i)$):

$$\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}(i)} \exp(e_{ik})}$$

(c) multi-head attention (K heads), then aggregate

K independent heads, params $W^{(k)}, a^{(k)}$



$$h'_i = \sigma \left(\sum_{j \in \mathcal{N}(i)} \alpha_{ij} Wh_j \right)$$

per head k , then concat $\parallel_{k=1}^K$ (hidden) or average (output)